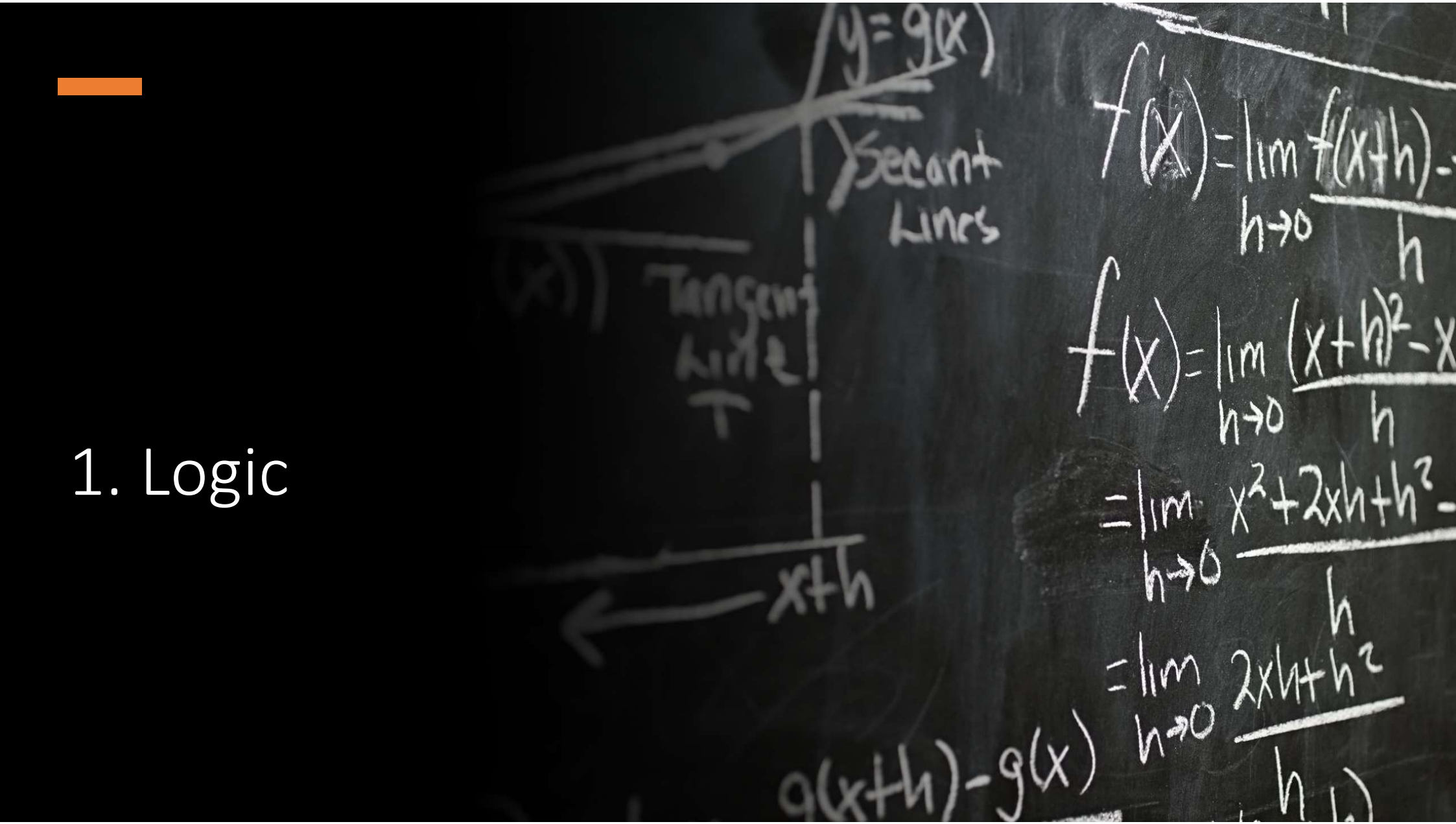
Week 0: LOGIC and MATHEMATICAL PROOF

Must-know technique in proving skills

1. Logic


y=g(x)
Secant Lines
Tangent Line
x+h
f'(x)=lim h→0 f(x+h)-
h
f(x)=lim h→0 (x+h)²-x
h
=lim h→0 x²+2xh+h²-
h
=lim h→0 2xh+h²
g(x+h)-g(x)

Statements

A statement, or proposition, is a sentence which is either true or false, but not both.

For example, 'It is now 4.30pm' is a statement but 'What is the time?' is not a statement.

Statements, or propositions, can be denoted by lower case letters p, q, r, ...

Examples of statements:

p: For all real values of x, $\int x dx = \frac{1}{2}x^2 + \text{constant}$. (True)

q: For any two positive real numbers a and b, $a - b = b - a$. (False)

r: For all real values of x, $\frac{d}{dx}\sin x = \cos x$. (True)

s: Any quadratic equation $ax^2 + bx + c = 0$ in one variable has two real roots if $b^2 - 4ac > 0$. (False)

Open statement

A declarative sentence is an *open statement* if

(1) it contains one or more variables, and

(2) it is not a statement, but

(3) it becomes a statement when the variables in it are replaced by certain allowable choices

Example (of open statement vs statement)

Sentence A: Let $x \in \mathbb{Z}$. Then $x^2 = 1$.

When $x = 1$, then it is true.

When $x = 0$, then it is false.

Sentence A is an open statement.

Sentence B: Let $x \in \mathbb{Z}$. Then $x^2 = 2$.

Sentence B is a statement. (Why?)

Consider for example function $\ln x$. We know that this function is defined for real positive values of x but not for negative values (or even zero) of x.

Now consider the sentence:

$$\int \tfrac{1}{x}\,dx = \ln x + \text{constant}.$$

If $x > 0$, then it is true.

But if $x \leq 0$, then it is false.

Taken out of context, $\int \tfrac{1}{x}\,dx = \ln x + \text{constant}$ cannot be said to be either right or wrong. The sentence is an open statement.

Definition

Mathematical objects come into existence by *definitions*. A mathematical definition must be absolutely clear. Definitions are usually followed by *examples* (like we have seen always) to further elaborate or explain the meaning.

Example: Divisibility

Definition: Let a and b be integers. Then a is divisible by b (or b divides a; or b is a *factor* of a; or b is a *divisor* of a) if there is an integer c such that $bc = a$. The notation for this is $b \mid a$.

Example: $4 \mid 12$ (4 divides 12) since $12 = 4 \times 3$ but 5 does not divide 12 since there exist no integer n such that $5n = 12$.

Example: Prime number

Definition: An integer $p > 1$ is *prime* if all of its positive divisors are 1 and p only.

Example: 11 is prime because it satisfies both conditions in the definition of prime. First $11 > 1$, and second, the only positive divisors of 11 are 1 and 11 itself. However, 12 is not prime because it has a positive divisor other than 1 and itself; for example, 3 divides 12 but $1 < 3 < 12$.

Remark

- The "if" statement in definition should be treated as "if and only if" (we will study these terms later) unless otherwise stated.
- However, for theorems, propositions, lemmas, facts, corollaries (in general all mathematical results), "if" and "if and only if" are NOT the same at all. So, you must pay attention to this.

Exercise 1

Try the exercise yourself and discuss with your peers

Exercise 1. State whether the following sentences are

(a) true for all real values of x,

(b) true for at least one real values of x, but not all x,

(c) true for no real values of x.

1. $x + 1 = 3$
2. $(x+1)^2 = x^2 + 2x + 1$
3. $x^2 + x + 1 = 0$
4. $(x-1)(x-2) \geq 0$
5. $(x-1)^2 \geq 0$
6. $\begin{pmatrix} x & 1 \\ 0 & 4 \end{pmatrix}$ has an inverse
7. $\cos x > 0$
8. $\begin{pmatrix} x & 1 \\ 0 & 4 \end{pmatrix}$ has no inverse

Which of the following statements are true? (Consider x to be a real number).

9. There is at least one value of x for which $x^2 + 1 = 0$.

10. There are no values of x for which $x^2 + x - 1 = 0$.

11. For all values of x, $\sin^2 x + \cos^2 x = 1$.

12. For at least one value of x, $\sin^2 x - \cos^2 x = 1$.

Answers: 1. b, 2. a, 3. c, 4. b, 5. a, 6. b, 7. b, 8. b, 9. false, 10. false, 11. true, 12. true.

Remark: We write $\equiv$ (instead of $=$) in the case where we have the equation is true for all real numbers, for example $(x+1)^2 \equiv x^2 + 2x + 1$, $\sin^2 x + \cos^2 x \equiv 1$.

Negation

Consider the statement p: I have a headache.

The statement 'I do not have a headache' is called the *negation* of p and is denoted by $\sim p$ (read as not p).

Examples:

a : It is snowing.	$\sim a$: It is not snowing.
b : All Chinese have black hair.	$\sim b$: It is not true to say that all Chinese have black hair.
c : 79 is a prime number.	$\sim c$: 79 is not a prime number.

d : For all real values of x, $x^2 + 1 > 0$.

$\sim d$: It is not true to say that for all real values of x, $x^2 + 1 > 0$.

Remark: We can rewrite d as $\forall x \in \mathbb{R}$, $x^2 + 1 > 0$; or $x^2 + 1 > 0 \ \forall x \in \mathbb{R}$.

We can write $\sim d$: $\exists x \in \mathbb{R}$ such that $x^2 + 1 \not> 0$; or $\exists x \in \mathbb{R} \ni x^2 + 1 \leq 0$.

Looking at the statements and their negations above we see that either the given statement, or its negation, is true. In fact, for any statement p,

if p is true then $\sim p$ is false;
if p is false then $\sim p$ is true.

Consider again the statement b : All Chinese have black hair. (False)

Now consider the following attempts at the statement $\sim b$:

b_1 : All Chinese do not have black hair.

b_2 : No Chinese has black hair.

b_3 : At least one Chinese does not have black hair.

On closer examination of $\sim b$: 'It is not true to say that all Chinese have black hair', we see that $\sim b \equiv b_3$ ($\sim b$ is equivalent to b_3, $\equiv$ is the symbol for equivalent). In other words, the negation of b can be written as

'At least one Chinese does not have black hair.'

Or 'Not all Chinese have black hair'.

Consider a similar mathematical statement,

'For all real values of x, $(x-1)(x-2) > 0$' (false)

The negation of the statement is

'At least one real value of x, $(x-1)(x-2) \leq 0$' (true)

But <u>not</u> as

'For all real values of x, $(x-1)(x-2) \leq 0$' (not true).

Exercise 2

Try the exercise yourself and discuss with your peers

Exercise 2. State which of the statements b, c, d, e is/are the negation of statement a.

1. a: θ is any positive angle.

   b: θ is any negative angle.

   c: There is at least one value of θ which is not positive.

2. a: 2 is a positive integer.

   b: 2 is not a positive integer.

   c: 2 is a negative integer.

3. a: Marmalade is made from oranges.

   b: Marmalade is not made from oranges.

   c: There is at least one variety of marmalade that is not made from oranges.

4. a: At least one make of car does not have four wheels.

   b: All cars have four wheels.

   c: All cars do not have four wheels.

   d: At least one make of car doe have dour wheels.

5. a: At quadratic equations in one variable have two real roots.

   b: There is no quadratic equation in one variable that has two real roots.

   c: At least one quadratic equation in one variable does not have two real roots.

   d: At least one quadratic equation in one variable has two real roots.

6. a: There is no real value of x for which x^2 is negative.

   b: $x^2 \geq 0$ for all real values of x.

   c: There is no real value of x for which x^2 is positive.

   d: x^2 is negative for at least one real value of x.

7. *a*: $f(x) > x$ for all real values of $x > 1$.

   b: $f(x) \not> x$ for all real values of $x > 1$.

   c: $f(x) \leq x$ for at least one real values of $x \leq 1$.

   d: $f(x) \leq x$ for at least one real values of $x > 1$.

8. *a*: For all integral values of n, $f(n) > n$.

   b: $f(n) \leq n$ for all integral values of n.

   c: $f(n) > n$ for no integral values of n.

   d: $f(n) < n$ for at least one integral values of n.

   e: $f(n) \leq n$ for at least one integral values of n.

Answers: 1. c, 2. b, 3. c, 4. b, 5. c, 6. d, 7. d, 8. e.

Conditional statements

For all real values of x, consider the following statement:

$$\text{If } x = 3 \text{ then } x^2 = 9 \text{ (is true).}$$

This is known as an *implication*, or *conditional statement*, which we write as

$$x = 3 \Rightarrow x^2 = 9\,, \text{ or } a \to b\,.$$

There are several ways of writing the linguistic equivalent of $a \rightarrow b$, e.g.,

$$x = 3 \text{ implies that } x^2 = 9$$

$$x = 3 \text{ therefore } x^2 = 9$$

$$x = 3 \text{ is sufficient for } x^2 = 9$$

$$x = 3 \text{ only if } x^2 = 9$$

In the conditional statement $a \rightarrow b$, a is called the hypothesis and b is called the conclusion.

When this is not the case and an implication involves a variable, we must know whether the variable can take *all* real values, or only *some* real values, before we can determine the truth of the implication.

For example, $x^2 = 4 \Rightarrow x = 2$ is not true within the context given at the beginning of the section but would be true in a context where $x > 0$.

Exercise 3

Try the exercise yourself and discuss with your peers

Exercise 3. All variables in this exercise can take any real value. Determine which of the following implications are true and which are false.

1. $(x+1)(x-2)=0 \Rightarrow x=-1$ or 2.
2. $x^2=16 \Rightarrow x=4$.
3. $f(x)\equiv(x+2)^2 \Rightarrow f(x)\geq 0$.
4. $\sin\theta=0 \Rightarrow \theta=0$.
5. $ax^2+bx+c=0$ has real roots $\Rightarrow b^2-4ac\geq 0$.

6. $\frac{dy}{dx} = 2x \Rightarrow y = x^2$.

7. $(x+1)(x-2) = 1 \Rightarrow x+1 = 1$ or $x-2 = 1$.

8. $\cos\alpha \cos\alpha = 1 \Rightarrow \cos 2\alpha = 1$.

9. $\sin\alpha \cos\alpha = 1 \Rightarrow \sin 2\alpha = \frac{1}{2}$.

10. $\cos\theta = 0 \Rightarrow \sin\theta = \pm 1$.

Answers. 1. True, 2. False, 3. T, 4. F, 5. T, 6. F, 7. F, 8. T, 9. F, 10. T

Converse of a conditional statement

We have seen that, when a statement involves a variable, it is important to know what values that variable can take. For the rest of this chapter, it is to be assumed that any variable is free to take all real values *unless otherwise stated.*

If in the implication $x = 3 \Rightarrow x + 1 = 4$, we reverse the arrow,

we get $x = 3 \Leftarrow x + 1 = 4$.

(Read as if $x + 1 = 4$ then $x = 3$).

Other ways to phrase it:

$x = 3$ is implied by $x + 1 = 4$.

$x = 3$ if $x + 1 = 4$.

$x = 3$ because $x + 1 = 4$.

$x = 3$ is necessary for $x + 1 = 4$.

In general, $q \rightarrow p$ is the *converse* of $p \rightarrow q$.

In the previous example, we have both implication and its converse are true. But this is not true in general!

Consider $x = 3 \Rightarrow x^2 = 9$ is true <u>but</u> $x = 3 \Leftarrow x^2 = 9$ is false because x can take -3.

Thus, a statement and its converse are not necessarily both true, nor both false.

Hence, it is unsound to argue from a statement to its converse, e.g., from

‘If this lecturer is from HWUM, then he is extremely brilliant’ to

‘If this lecturer is extremely brilliant, then he is from HWUM’.

This is a common error of reasoning!

The negation of a conditional statement

The negation of a conditional statement $p \to q$ is $\sim (p \to q)$.

$\sim (p \to q) \equiv p \wedge \sim q$, $\wedge$ denotes and (conjunction of two statements)

Now consider $x^2 = 9 \Rightarrow x = 3$ (false). The negation of this statement is

$$x^2 = 9 \wedge \sim (x = 3), \text{ or equivalently}$$

$$x^2 = 9 \wedge x \neq 3 \text{ (true).}$$

The statement $x^2 = 9 \wedge x \neq 3$ is true because there is *one* value of $x = -3$ that satisfies the statement it.

Consider another example, suppose $p \rightarrow q$ is

'If all the cars in this showroom that are made in Malaysia then they are all good cars'.

Then $\sim (p \rightarrow q) \equiv p \wedge \sim q$ is

'Of all the cars in this showroom that are made in Malaysia, there is at least one which is not good'.

Biconditional statements

Consider two statements a: $x = 3$, and b: $x + 1 = 4$.

Again, we do not know the truth of a or b, but we can say

$$\text{If } x = 3 \text{ then } x + 1 = 4 \textit{ and } \text{if } x + 1 = 4 \text{ then } x = 3.$$

That is, $a \to b \wedge b \to a$. This is known as a *biconditional statement* and written as

$$a \leftrightarrow b \equiv a \to b \wedge b \to a \text{ means } a \text{ if and only if } b$$

Some other ways to phrase $a \leftrightarrow b$ are

‘$x = 3$ implies and is implied by $x + 1 = 4$’;

‘$x = 3$ if and only if by $x + 1 = 4$’; (*iff* as abbreviation)

‘$x = 3$ is necessary and sufficient for $x + 1 = 4$’.

$a \leftrightarrow b$ is true only if the true statements are obtained for both directions.

For example,	$\sin u = 0 \Rightarrow \tan u = 0$	is true
	$\sin u = 0 \Leftarrow \tan u = 0$	is true
Hence	$\sin u = 0 \Leftrightarrow \tan u = 0$	is true.
But	$\cos u = 1 \Rightarrow \sin u = 0$	is true
	$\cos u = 1 \Leftarrow \sin u = 0$	is false
So	$\cos u = 1 \Leftrightarrow \sin u = 0$	is false.

Contrapositive statements

Consider the statements $\begin{cases} a : x = 3 \\ b : x^2 = 9 \end{cases}$ and their negations, $\begin{cases} \sim a : x \neq 3 \\ \sim b : x^2 \neq 9 \end{cases}$.

The true conditional relationship between a and b is $a \rightarrow b$,

whereas the true conditional relationship between $\sim a$ and $\sim b$ is $\sim b \rightarrow \sim a$.

$\sim b \rightarrow \sim a$ is said to be the *contrapositive statement* of $a \rightarrow b$ and the important relationship between them is given:

$$a \rightarrow b \equiv \sim b \rightarrow \sim a \, .$$

Remark: Proving by its contrapositive statement is very useful in the time when proving the original conditional statement is difficult. We will encounter prove by contrapositive in many proofs in future.

Exercise 4

Try the exercise yourself and discuss with your peers

Exercise 4. In Questions 1 – 9, insert the correct conditional symbol between them (i.e., $p \to q$, $q \to p$, or $p \leftrightarrow q$). Assume that the variables can take any real value.

	p	q
1.	$x + 2 = 4$;	$x = 2$.
2.	$x^2 = 4$;	$x = 2$.
3.	$\theta = \frac{\pi}{4}$;	$\sin\theta = \frac{\sqrt{2}}{2}$.
4.	$P(x, y)$ is a point on the circle, centre O, radius 2;	$x^2 + y^2 = 4$

5.	$\tan\theta = \pm 1$;	$\sin\theta = -\frac{\sqrt{2}}{2}$.
6.	$\frac{x-1}{(x+2)^2} = 0$;	$x = 1$.
7.	x^2 is even;	x is even.
8.	x is rational;	x^2 is rational.
9.	$x^2 + ax + b = 0$ where a and b are real;	the product of the roots is real.

10. Write the (a) converse, (b) negation, (c) inverse, (d) contrapositive of 'if $x^2 = -1$ then x is not real'.

11. Determine which of the following is equivalent to the statement

$$f(x) \equiv x^2 + 5 \Rightarrow f(x) \not< 5 \ \forall x \in \mathbb{R}$$

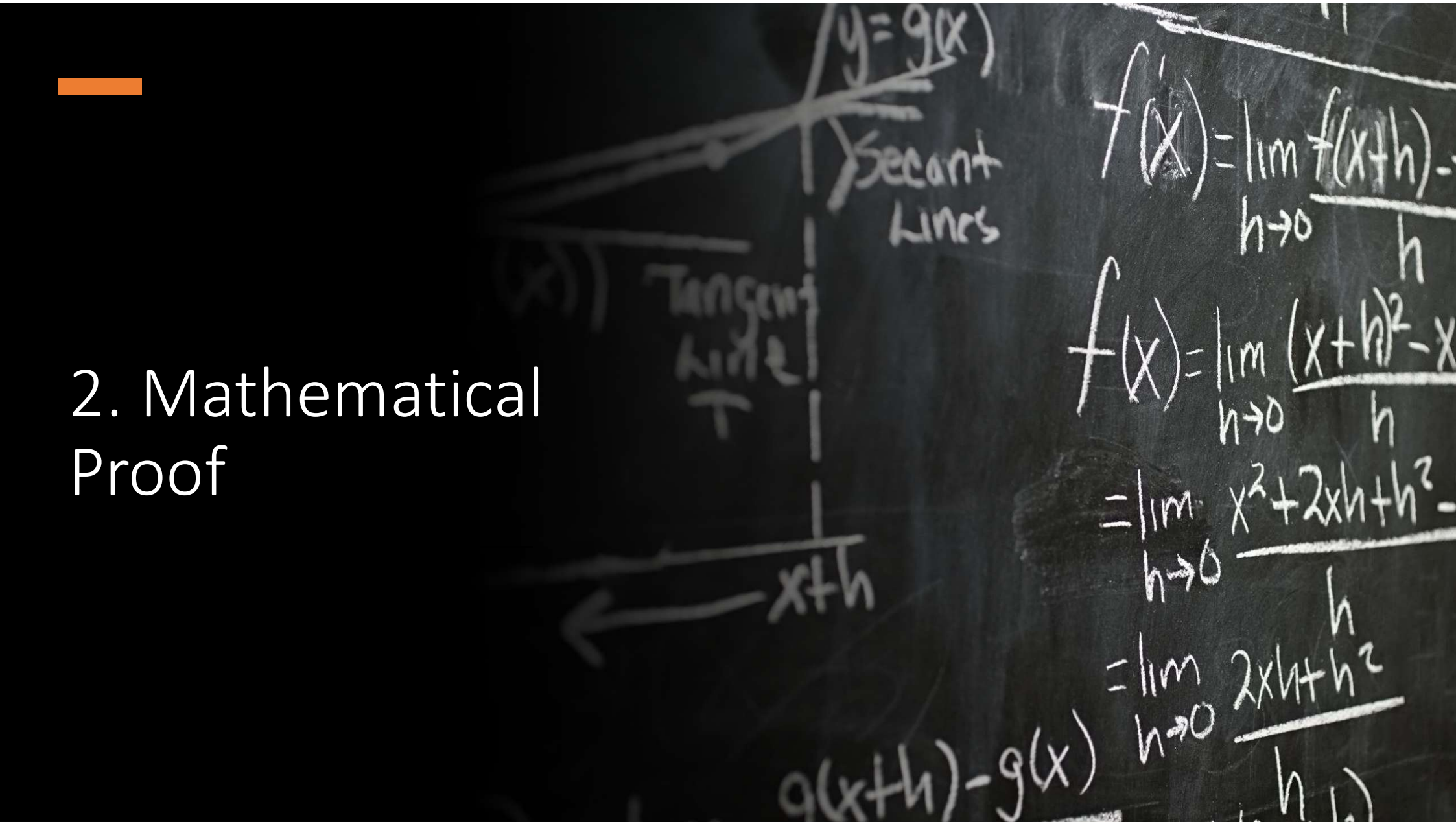
(a) $\exists x \in \mathbb{R} \ni f(x) = x^2 + 5$ and $f(x) \geq 5$. (b) $f(x) \not\equiv x^2 + 5 \Rightarrow f(x) < 5$

(c) $\exists x \in \mathbb{R} \ni f(x) < 5 \Rightarrow f(x) \not\equiv x^2 + 5$

Answers. 1. $p \leftrightarrow q$, 2. $p \leftarrow q$, 3. $p \rightarrow q$, 4. $p \leftrightarrow q$, 5. $p \leftarrow q$, 6. $p \leftrightarrow q$, 7. $p \leftarrow q$

8. $p \rightarrow q$, 9. $p \rightarrow q$, 10. (a) If x has no real value then $x^2 = -1$, (b) $x^2 = -1$ and x is real, (c) If $x^2 \neq -1$ then x is real, (d) If x is real then $x^2 \neq -1$. 11. (c)

2. Mathematical Proof



Direct proof: To prove $p \to q$ is true, start with p then deduce

$$p \equiv p_1 \to p_2 \to p_3 \to \cdots \to p_n \equiv q \text{ so that } p \to q \text{ is true.}$$

Example. Prove that sum of two even integers is even.

Proof. An integer a is even if a can be expressed as $2k$ for some integer k. Let x and y be two even integers. Then we may write $x = 2m$, $y = 2n$ for some integers m and n. So, $x + y = 2m + 2n = 2(m + n) \in 2\mathbb{Z}$ is even.

Proof by induction: Let p_n be a statement involving n, where n is any positive integer.

1. Prove that $p_k \rightarrow p_{k+1}$ is true.
2. Prove that p_1 is true.
3. Combine steps 1 and 2 to show that p_r is true for all $r = 1, 2, \ldots, n$.

The third step is usually expressed in the form 'therefore, by induction, the result is correct for all positive integral values of n'.

Example. Prove that $9^n - 1$ is a multiple of 8 for all positive integers n.

Proof. Step 1: Suppose $9^k - 1 = 8M$ for some $n = k > 1$ and for some integer M. Then

$$9^{k+1} - 1 = 9(9^k) - 1 = 9(9^k - 1) + 8 = 9(8M) + 8 = 8(9M + 1),$$

which shows that $9^k - 1$ is a multiple of 8 implies that $9^{k+1} - 1$ is a multiple of 8.

Step 2: Obviously $9^1 - 1 = 8$ is a multiple of 8.

Step 3: By induction, the result is correct for all positive integer values of n.

Exercise 5

Try the exercise yourself and discuss with your peers

Exercise 5. Prove by induction the following statements is true for all positive integer values of n.

1. $\frac{d^n}{d\theta^n}(\sin a\theta) = a^n \sin\left(a\theta + \frac{n\pi}{2}\right)$

2. $\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}$

3. $\left|\sum_{i=1}^{n} z_i\right| \leq \sum_{i=1}^{n} \left|z_i\right|, n > 1$

4. $(1+x)^n \equiv \sum_{r=0}^{n} {}^nC_r x^r, {}^nC_r = \frac{n!}{r!(n-r)!}$

5. $2^{n+2} + 3^{2n+1}$ is divisible by 7

6. $2^n > 2n$, $n > 2$

7. $\int x^n dx = \frac{x^{n+1}}{n+1} +$ constant (use integration by parts)

8. $A = \begin{pmatrix} 2 & a \\ 0 & 1 \end{pmatrix} \Rightarrow A^n = \begin{pmatrix} 2^n & (2^n - 1)a \\ 0 & 1 \end{pmatrix}$.

Indirect proof (contrapositive)

Proof by contrapositive: When it is difficult to prove $p \rightarrow q$, a direct proof of the contrapositive statement $\sim q \rightarrow \sim p$ can be used.

Example. Prove that if n is a natural number such that n^2 is even, then n is even.

Proof. The contrapositive of the statement 'n^2 is even, then n is even' is

$$n \text{ is odd then } n^2 \text{ is odd.}$$

Now n is odd, meaning $n = 2k+1$ for some integer k. Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 \text{ is odd.}$$

Hence, if n is odd then n^2 is odd. Subsequently, if n^2 is even, then n is even.

Proof by contradiction

A *contradiction* is a statement that is always false. For example, '$x = 2$ and $x \neq 2$' is false. So any statement of the form $q \wedge \sim q$ is false.

Proof by contradiction: To verify a statement p we start with $\sim p$ and deduce a statement of the form $q \wedge \sim q$. As this is false we can argue that $\sim p$ is false, and hence p is true.

Example. There are no largest natural number. Hence, there are infinitely many natural numbers.

Proof. The set of natural number is $\mathbb{N} = \{1, 2, 3, ...\}$. Suppose that there indeed exists the largest natural number M. Then we consider the number $M + 1$, certainly

$$M + 1 > M \text{ and } M + 1 \text{ is a natural number.}$$

Thus, we get a contradiction (as we say that M is the largest natural number).

Example. Prove that $\sqrt{2}$ is not a rational number.

Proof. Assume that $\sqrt{2} \in \mathbb{Q}$ and thus write $\sqrt{2} = \frac{m}{n}$ where $m, n \in \mathbb{Z}^+$ with no common factor. Then

$$
\begin{aligned}
& 2n^2 = m^2 \\
\Rightarrow\ & m^2 \quad \text{is even} \\
\Rightarrow\ & m = 2k, k \in \mathbb{Z} \\
\Rightarrow\ & 2n^2 = 4k^2 \\
\Rightarrow\ & n^2 \quad \text{is even} \\
\Rightarrow\ & n \quad \text{is even}
\end{aligned}
$$

(Here, we are using the result from example we showed earlier in proof by contrapositive, i.e., n^2 is even implies that n is even.) Hence, we have a contradiction to the fact that m and n are integers with no common factor. So $\sqrt{2}$ is irrational.

Counter example – to disprove a statement

Disprove by counter example: To prove that a statement is false, all that we have to do is to produce just *one case* to show that the statement is indeed false.

Example. Show that $\left|x\right| = \left|y\right|$ then $x = y$.

Disproof by counter example: Take $x = 1 = -y$. Then $\left|x\right| = \left|y\right|$ but $x \neq y$.

Exercise 6

Try the exercise yourself and discuss with your peers

Exercise 6. For Question 1 – 3, prove each of the statements by using its contrapositive.

1. If n is a perfect number, then n is not a prime number.
2. If n is a natural number such that n^2 is odd, then n is odd.
3. $(x-2)(1-x) > 0 \Rightarrow 1 < x < 2$

In Question 4 – 6, use proof by contradiction.

4. $\sqrt{3}$ is irrational.
5. There are infinitely many primes.
6. For all $x > 0$, $x + \frac{1}{x} \geq 2$.

In Question 7 – 9, find a counter example to disprove the statement.

7. For any two real numbers a and b, $\frac{a}{b} = \frac{b}{a}$.
8. For any two vectors $\underset{\sim}{a} \times \underset{\sim}{b} = \underset{\sim}{b} \times \underset{\sim}{a}$.
9. The sum of two unequal, positive, irrational numbers is irrational.

Existence and uniqueness proof

An *existential quantification* is a type of quantifier, termed as 'there exists', 'there is at least one', or 'for some'. It is usually denoted by the logical operator symbol $\exists$, which, when used together with a predicate variable, is called an existential quantifier (for example $\exists x \in \mathbb{R}$). Existential quantification is distinct from *universal quantifier* ('for all' denoted by $\forall$), which asserts that the property or relation holds for *all* members of the domain.

The term *uniqueness* refers to the property of being the one and only object satisfying a certain condition. This sort of quantification is known as uniqueness quantification or unique *existential* quantification, and is denoted with the symbols '$\exists!$' or '$\exists_{=1}$'. For example, the formal statement $\exists! n \in \mathbb{N} \ni n - 2 = 4$ (read as 'there is exactly one natural number n such that $n - 2 = 4$'.

Proof of existence: Provide an example for which it will satisfy the statement.

Example: Prove that $\exists n \in \mathbb{N}$ such that n is equal to the sum of its positive divisors, except n itself.

Proof. Let $n = 6$. The positive divisors of 6 (except itself) are 1, 2, and 3 and

$$1 + 2 + 3 = 6\,.$$

Remark: The numbers with such property are called *perfect numbers*.

Proving uniqueness: First prove the existence of the entity with the desired condition, and then to prove that any two such entities (say, s and t) must be equal to each other (i.e. $s = t$).

Example. Show that $x + 2 = 5$ has unique solution.

Proof. We first show that the equation has at least one solution exists, namely 3; it is easily verified as follow: when $x = 3$, $x + 2 = 5$.

To establish the uniqueness of the solution, one would then proceed by assuming that there are two solutions, namely s and t, satisfying $x+2=5$, i.e., $\begin{matrix} s+2=5 \\ t+2=5 \end{matrix}$. Hence, $s+2=t+2 \Rightarrow s=t$.

Biconditional proof: To prove a statement 'p if and only if q', we prove

$$p \to q \text{ is true } \underline{\text{and}} \ q \to p \text{ is true.}$$

Remark: One can make use of method learned before, i.e., contrapositive, contradiction, etc. in one or both directions.

We write $\Rightarrow$) when we are proving $p \to q$ and write $\Leftarrow$) when proving $q \to p$.

Example. Let n be an integer. Prove that n is even if and only if n^2 is even.

Proof. $\Rightarrow$) When n is even, $n = 2k$ for some integer k. Then $n^2 = 4k^2 = 2(2k^2)$ is even.

$\Leftarrow$) We have proven this using contrapositive.